ThunderWorld: where to spend A/W repair, and what weight fusion can save

RTX 5090 · 11 September 2026 · experimental implementation, not a default policy

The evidence supports **module- and task-dependent repair allocation** and **shape-dependent kernel selection**. It does not support dropping weight repair everywhere, or treating two FP4 planes as one FP4 multiply at unchanged precision.

1. Keep the algebra and the execution model separate

Using column-vector notation, the first-order approximation is

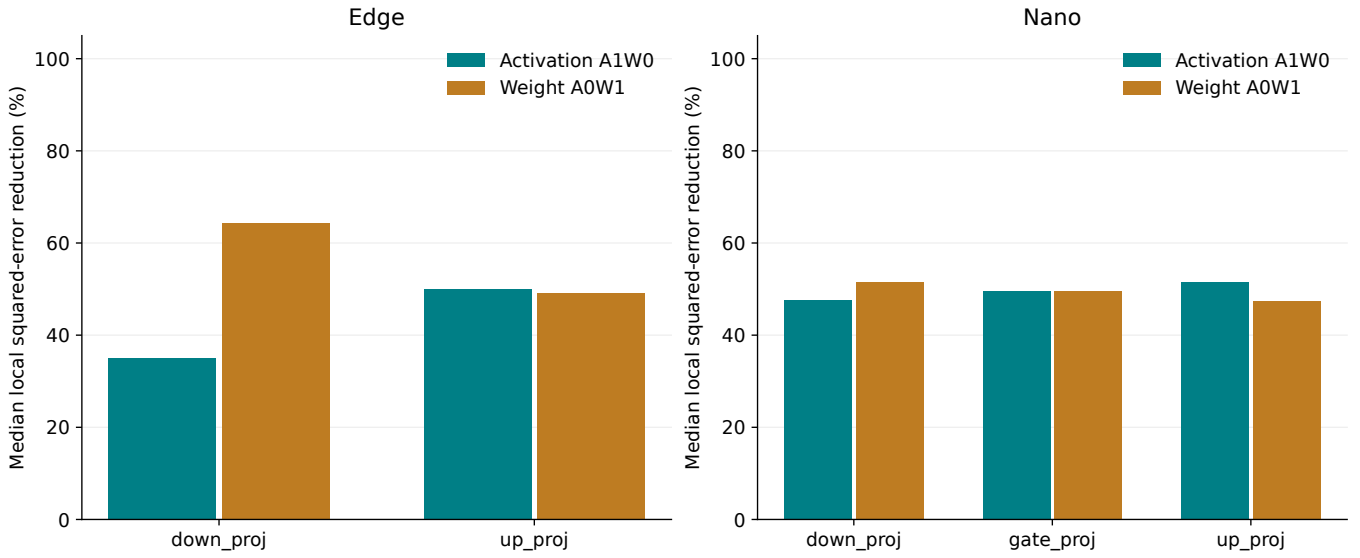
$Y \approx W_0 A_0 + W_0 A_1 + W_1 A_0$.

The repository uses the transposed row-vector convention: $A_0 W_0 + A_1 W_0 + A_0 W_1$. The cross term is omitted in this study. The native FFN already fuses the base and selected residual products into **one kernel and one FP32 accumulator**, with a final BF16 store. Avoid claiming that the starting point launches three independent GEMMs.

Component	What is offline?	Residual products
FFN; QKV/output projections	Weight planes and their scales	Activation repair and weight repair
Attention QK	Neither Q nor K is an offline weight	$Q_1 K_0^\top$, $Q_0 K_1^\top$
Attention PV	Neither probabilities nor V is an offline weight	$P_1 V_0$, $P_0 V_1$

Here, attention “A-only” means the current implementation's $Q_1 K_0^\top$ and $P_1 V_0$ repairs. Their budgets remain coupled. It does **not** imply that offline projection-weight residuals have been repaired.

2. FFN: weight repair is particularly valuable in down projections



Model / projection	Activation repair: median error reduction	Weight repair: median error reduction	Probe units
Edge / up	50.0%	49.0%	336
Edge / down	35.1%	64.3%	336
Nano / up	51.6%	47.4%	720
Nano / gate	49.5%	49.6%	720
Nano / down	47.5%	51.5%	720

These are reductions of **normalized squared local output error** relative to base FP4, from existing seed-7101 calibration probes. They are not percentage reductions of final video error. Captured rows, heads, layers and phases are not independent task samples. Probe scales/support can differ from the complete trajectory.

The split candidate therefore uses **A repair in up/gate and W repair in down**, at the same two-product arithmetic budget as all-A or all-W. Gate's near tie is not evidence for a strong universal preference. The complete-trajectory screen below is the necessary check on this local heuristic.

The archived complete-trajectory ablation (seeds 6103/6104, BF16 attention fixed) supplies the base and dual-repair controls:

FFN repair	Edge I2V SSIM ↑	Nano I2V SSIM ↑	Nano audio relative L2 ↓
None	0.534	0.923	1.150
A only	0.533	0.956	0.577
W only	0.573	0.922	0.393

FFN repair	Edge I2V SSIM ↑	Nano I2V SSIM ↑	Nano audio relative L2 ↓
A + W	0.692	0.960	0.049
A + W + cross	0.771	0.964	0.077

SSIM is the two-seed mean on `robot_table`; audio is the worst decoded-waveform relative L2 on `av_pottery`, both against BF16. These are different tasks and are not averaged into one score. W repair remains important for audio and for the dual-repair quality target. Cross repair is not uniformly beneficial. Full values, action metrics and source hashes are in `term_quality.csv` and `term_sources.json`.

3. Attention: use features as a prior, then calibrate repair benefit

The implemented candidate compiles development-only BF16 observations into head × layer × step-phase directed density priors. It accounts for actual AR, clean/noisy vision, sound and action token layouts, including mixed-role and tail tiles. A global ranking keeps exactly half of the padded 128×128 repair tiles; **every base tile remains active**. The runtime retains a joint streaming softmax and computes fresh values at each step.

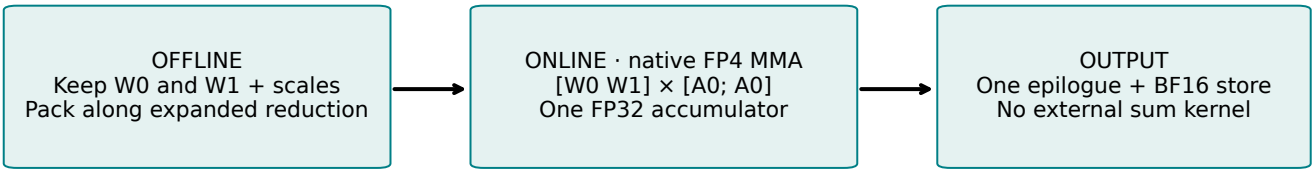
The comparator uses current attention mass × V-energy at the same total half-tile quota. Both repair QK and PV using terms=2. This is a matched padded-tile budget, not a claim of exactly equal useful FLOPs at sequence tails.

Implemented	Still required before a production policy
Development-only priors, four step phases	Predict marginal repair benefit, not mass alone
Directed role-pair density, token-length normalization	Task-specific action/audio quality floors
Global head/tile ranking	Separate QK and PV budgets; compare K/V-side repair
No support search during repeated structured calls	Matched uniform/random controls; broader held-out tasks/seeds
Full base connectivity	Quality guardrails before dropping any repair family

AR–DM depth trends and modality asymmetry are **features**, not proof that a tile is safe to leave unrepaired. CFG branches are averaged in this prototype. A per-head scalar cannot alter within-head top-k ranking unless it also changes head quotas or participates in a global allocation; this implementation uses global ranking.

A practical allocator should estimate `benefit(t, tile) = L(without repair) − L(with repair)` and choose repair terms/tiles subject to measured latency and memory budgets. Estimate benefit on representative layer/step probes, then accept policies using full-trajectory video, action and audio checks. Non-additive errors mean that a greedy local score remains a candidate policy, not an optimality guarantee.

4. Offline weight merging: exact representation costs 2K



Exact residual representation still needs 2K products. A0 loads can alias the same buffer.

Requantizing W0 + W1 into one FP4 plane gives K work, but introduces new quantization error.

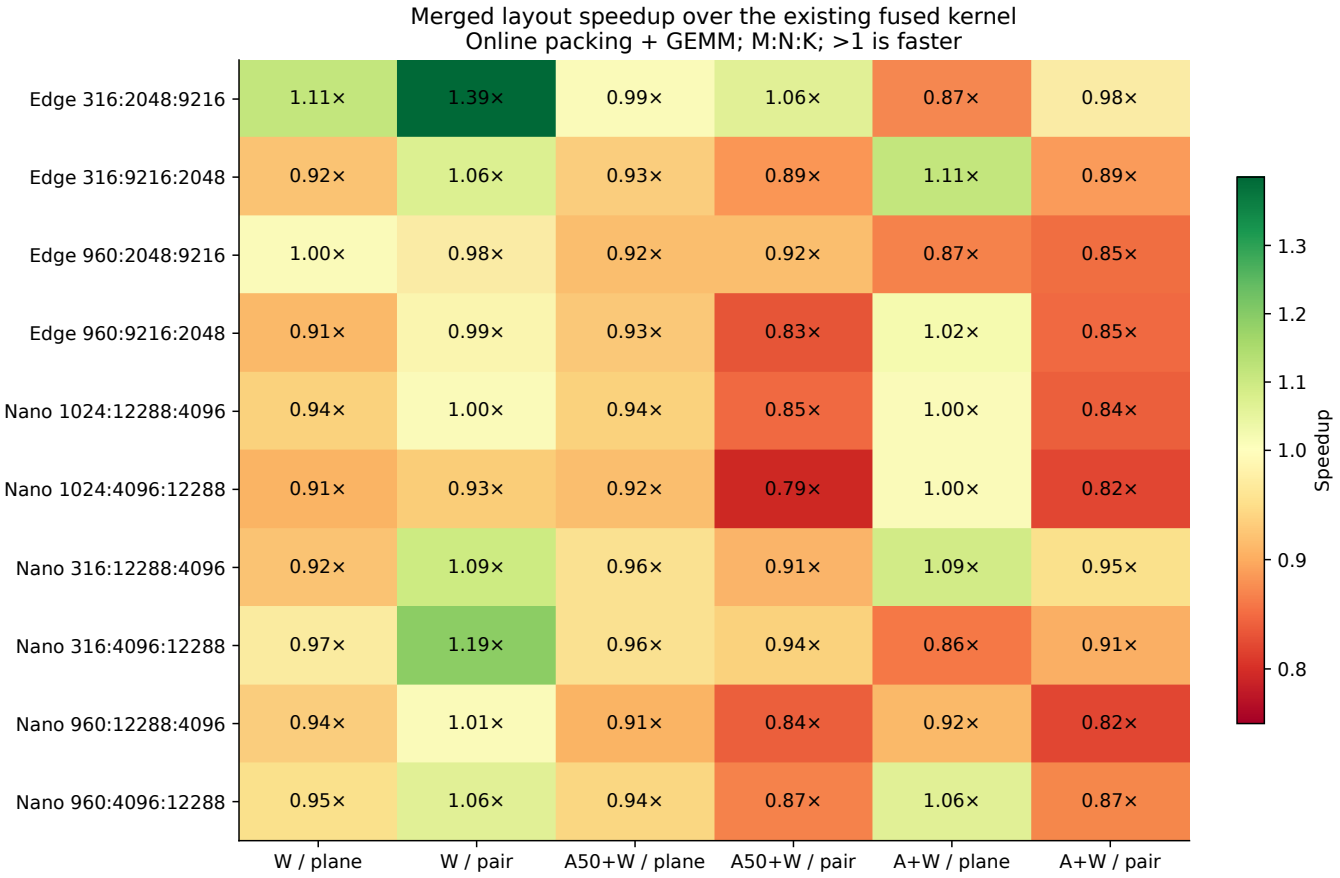
`(W0 + W1) A0 = [W0 W1] [A0; A0]` is exact before rounding. Keeping both FP4 planes allows a single GEMM reduction over **2K**, with activation loads aliasing the same buffer. No additional output-reduction kernel is necessary: the existing FP32 accumulator performs the sum.

In general, `W0 + W1` cannot be represented exactly by one E2M1 FP4 plane with one scale per block. For example, `[6, 0.5] + [0.5, 0] = [6.5, 0.5]` has positive-value ratio 13; one scaled FP4 plane's largest-to-smallest positive ratio is 12. Requantizing the sum restores K work but adds a new error. BF16 storage changes the arithmetic/storage trade-off and does not retain the two-plane FP4 representation.

Two layouts were implemented:

Layout	Reduction organization	Numerical consequence
`plane`	Base plane, then weight residual plane; A repair interleaved as before	Original accumulation order; all valid local comparisons bitwise equal
`pair`	Pair W0/W1 inside expanded K256 tiles; optional A repair afterward	Different summation order; maximum local relative L2 to original 0.000242

Both emit native block-scaled E2M1 MMA instructions. This verifies Tensor Core execution, **not saturation or achieved peak utilization**. Reduction-dimension residual concatenation also has related work; it is not claimed as a new standalone algebraic contribution. See Triton scaled dot, block-scaled matmul, and ARCQuant.

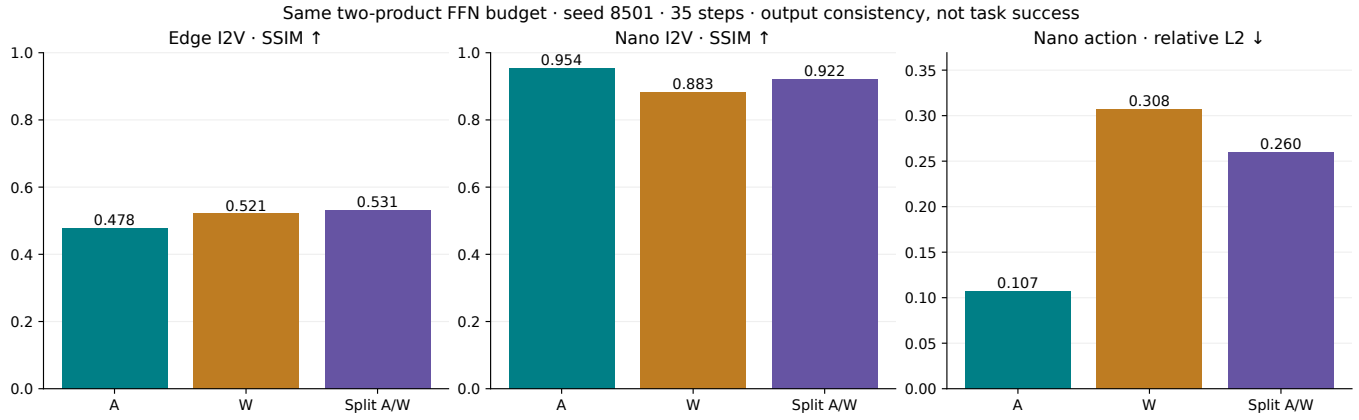


The benchmark covers **10 real model shapes × 3 repair tiers**, 530 valid numerical configurations and 10 shared-memory resource skips. Captured 64-row inputs are repeated to the actual M for timing; they do not add quality samples. Online packing is included; support selection is prepared before timing. Winners are retimed in five randomized CUDA Graph rounds against the existing tuned fused kernel.

The best paired W-only cases reach **1.39× on Edge** and **1.19× on Nano**. Many A50+W and full A+W shapes regress. Therefore this is an optional dispatch choice after shape-specific tuning, not a global replacement. The trajectory prototype retains fallback and merged weights simultaneously; its memory allocation is **not** deployment memory.

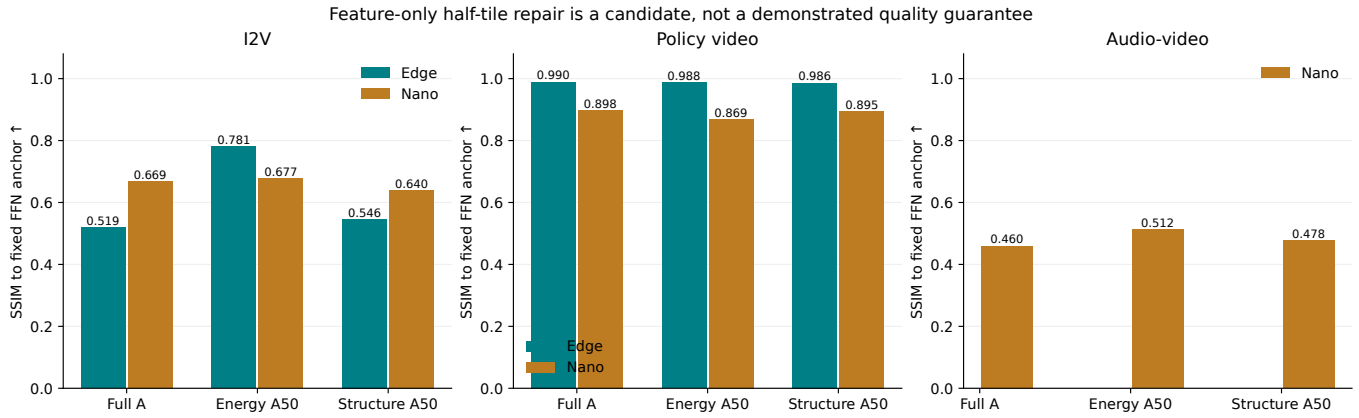
5. Fresh full-trajectory screen

The fresh screen contains **40 unique mode/task trajectories**: Edge I2V and policy, and Nano I2V, policy and audio-video, all at seed 8501 and 35 steps. This is one fresh seed on existing evaluation scenes, not a broad held-out benchmark. The initial Edge campaign stopped at the new attention wrapper's causal-AR guard; the guard was corrected to preserve the original causal path. Its 12 completed rows are retained with the failed campaign status. A separate completion campaign repeated four anchors, all verified bitwise identical, and supplied the remaining attention rows. These repeats are not extra samples.



Fixed BF16 attention; FFN variant	Edge I2V SSIM ↑	Edge action L2 ↓	Nano I2V SSIM ↑	Nano action L2 ↓	Nano waveform L2 ↓
A everywhere	0.478	0.277	0.954	0.107	0.352
W everywhere	0.521	0.031	0.883	0.308	0.478
A up/gate; W down	0.531	0.146	0.922	0.260	0.232

All columns compare to BF16; they are distinct tasks. The split improves Edge I2V and Nano audio, but regresses Nano I2V/action versus A-only. Even Edge action prefers all-W on this seed. **Do not install a universal down=W rule.** The local attribution is useful for proposing allocations, while the trajectory result decides acceptance.



For attention isolation, the FFN is fixed to A-only and SSIM is measured against that same FFN with BF16 attention. Nano I2V SSIM is **0.669 / 0.677 / 0.640** for full A repair / dynamic-energy A50 / structural A50. Nano decoded waveform relative L2 to the same FFN anchor is **2.743 / 2.548 / 2.783**. The structural prior does not consistently beat the matched dynamic selector, and full A repair itself is insufficient for the current high-quality target. A slightly better BF16 SSIM can hide larger perturbations relative to the fixed FFN anchor; both references are retained in the CSV.

A separate two-step dispatch audit confirms that **Edge I2V/policy and Nano policy/AV fall back to the original kernel**. The current policy input has M=416, whereas the tuning table's policy probe has M=316; Edge I2V and Nano AV have no selected winning W-only layout. Their bitwise-equal split/merged outputs therefore do not validate merged-kernel equivalence. Nano I2V (M=960) actually selects the paired layout. Nano I2V does not: video relative L2 between the two split variants is **0.1222**, and trajectory-sketch relative L2 is **0.1753**, despite BF16 SSIM

changing only from 0.9219 to 0.9202. Small local accumulation differences can amplify over denoising steps. The paired layout therefore needs a complete-trajectory acceptance gate; this run does not establish equivalence.

Generator request timings are recorded for diagnosis, without throughput claims: variants were not repeatedly randomized as end-to-end performance trials. Waveform decode ran separately after releasing the Transformer. No integrated audio-request latency is inferred by adding the two timings.

See `trajectory_screen.csv`, `trajectory_audit.json`, and `audio_screen.csv` for every configuration and its source. The failed initial Edge run lacks the wrapper's end-of-run hash metadata; the successful completion and Nano campaigns include it. Their actual pre-receipt wrapper is preserved under `source_snapshots`, and all saved output-array hashes were verified.

6. What changed, validation, and next implementation decisions

Experimental code is isolated in tools/repair_design. Production defaults remain unchanged. The study adds attribution analysis, two native merged-weight layouts, structured attention masks, development-only prior fitting, a process-local trajectory integration, benchmark search and independent numerical validation.

Compute Sanitizer memcheck completed with **0 errors**. GPU validation passed 12 FFN combinations including M=1/257 tails, an independently decoded FP4 reference, and CUDA Graph replay after input mutation. Full/empty external attention masks match the corresponding original paths bitwise, and mixed-role/tail masks obey exact quotas. The benchmark records resource failures explicitly rather than silently dropping them.

Decision	Recommended implementation
Allocate A versus W	Calibrate by projection, layer, phase and task; retain dual repair where its marginal benefit is high
Select attention tiles	Combine head/role/phase priors with measured repair sensitivity and protected task budgets
Reduce runtime overhead	Freeze or bucket masks/shapes for graph replay; keep optional merged layouts only when consistently faster
Promote a kernel/policy	Require actual dispatch receipts, matched end-to-end quality, repeated timing, and deployment memory accounting

The dispatch audit also checks all 1,056 phase/layer mask receipts across eight real role layouts. Calls include warmup, and these two-step runs are not quality samples. Policy M=416 requires its own tuning and full-trajectory verification before enabling a merged layout there.

No energy measurements or real robot task-success evaluations were added. Output agreement with BF16 is a consistency measure, not a complete perceptual or downstream-quality assessment.

Raw evidence and reproducible figures: results/repair_design_20260911. Prior two-seed term ablations, fresh one-seed trajectories and local kernel probes remain separate cohorts.